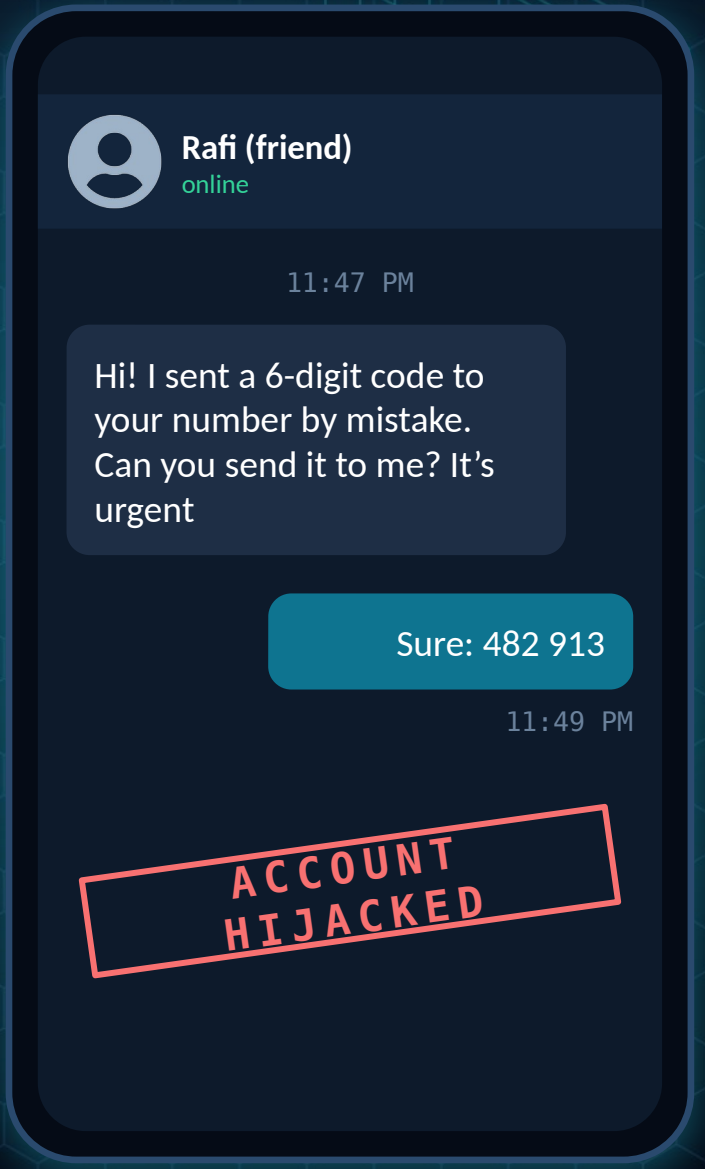
11:47 PM · DHAKA · AN ORDINARY WEEKNIGHT

One message. One night. Five doors.

Locked or Leaked? A security investigation of Facebook, Instagram, WhatsApp, Gmail and Telegram



MCCJ 525 · CYBERCRIMES AND SECURITY

[Member 1] · [Member 2] · [Member 3] · [Member 4] · [Member 5]

Group [__] | [Department], [University] | [Date]

Nusrat is a fictional composite based on common attack patterns in Bangladesh.

// 01 INCIDENT TIMELINE

In 81 minutes, one shared code cost Nusrat four accounts

11:49 PM

Shares the code. WhatsApp hijacked: no two-step PIN.

11:58 PM

Friends get “urgent” requests to send bKash money.

12:20 AM

Fake “copyright” link steals her reused password.

12:35 AM

Password reset via Gmail; recovery details swapped.

01:10 AM

Private DMs read and used to blackmail her.

SHE IS NOT ALONE // CYBERCRIME IN BANGLADESH

21.65%

of victims: account hacking, the #1 cybercrime

≈59%

of cybercrime victims are women

78.78%

of victims are aged 18–30, like us

12%

of victims ever report it to police

Sources: Cyber Crime Awareness Foundation (2024), Cybercrime Trends in Bangladesh, reported by Dhaka Tribune. Timeline is illustrative.

We retraced the attacker's path and tested every lock

DOOR 1
WhatsApp
11:49 PM
BREACHED

DOOR 2
Gmail
12:20 AM
BREACHED

DOOR 3
Facebook
12:35 AM
BREACHED

DOOR 4
Instagram
01:10 AM
BREACHED

DOOR 5
Telegram
MEANWHILE
AT RISK

THE FIVE LOCK TESTS // EACH SCORED 1 (WEAK) TO 5 (STRONG), MAXIMUM 25

Login protection

Can a stolen password alone get in?

Encryption by default

Can the company read my messages?

Threat alerts

Will it warn me of an attack?

Recovery safety

Can a hacker lock me out for good?

Privacy controls

Can strangers find and reach me?

WhatsApp: the strongest lock, opened from the inside

11:49 PM *The encryption never broke. Nusrat opened the door herself by sharing a six-digit code, and no two-step PIN stood in the way.*

BREACHED

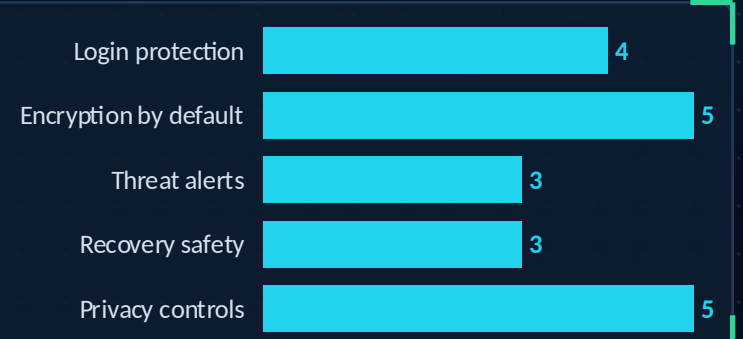
THE LOCKS THAT EXIST

- End-to-end encryption on every chat and call, by default
- Six-digit two-step verification PIN
- Passkey login and passkey-locked backups (Oct 2025)
- Chat lock, silence unknown callers, IP protection
- Strict Account Settings: one-tap lockdown (Jan 2026)

LOCK SCORE

20 /25

JOINT 1ST



HOW TO FIX THIS DOOR

A Bangla warning whenever a registration code is requested, and a mandatory two-step PIN.

Sources: WhatsApp (2026); Help Net Security (2025).

Gmail: the master key that opens every other door

12:20 AM *She typed her password into a fake "copyright violation" page. Same password as Gmail, no 2-Step Verification: the master key was gone.*

BREACHED

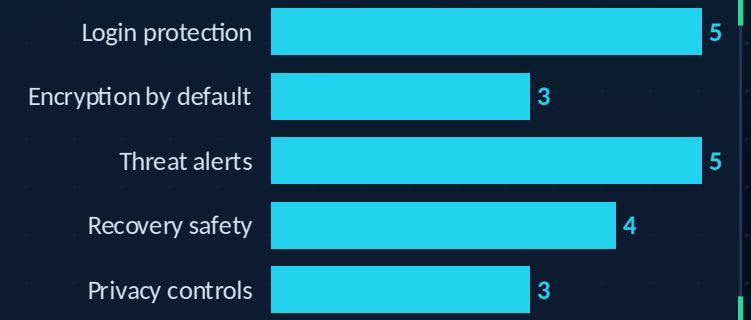
THE LOCKS THAT EXIST

- Passkeys as the default sign-in since Oct 2023
- 2-Step Verification and Security Checkup
- Critical alerts for suspicious sign-ins
- Filters block 99.9%+ of phishing and malware (Google)
- Advanced Protection Program for high-risk users

LOCK SCORE

20 /25

JOINT 1ST



HOW TO FIX THIS DOOR

Optional end-to-end encryption for personal mail, and lookalike warnings for bKash, Nagad and local banks.

Sources: Google (2023); Engadget (2025).

Facebook: strong locks, but the attacker changed them

12:35 AM *Using her Gmail, the attacker reset her Facebook password and swapped the recovery email and phone. By morning, Facebook believed he was Nusrat.*

BREACHED

THE LOCKS THAT EXIST

- Two-factor authentication: app, SMS or security key
- Login alerts and remote log-out
- Passkeys on mobile since June 2025
- Profile lock hides photos from strangers
- Messenger chats end-to-end encrypted by default



HOW TO FIX THIS DOOR

A 48-hour cool-down on recovery changes, with alerts to old contacts, plus Bangla-language human support.

Sources: Meta (2025); Crisan (2023).

Instagram: her private messages were never truly private

01:10 AM *Linked through Meta's Accounts Center, Instagram fell next. Her DMs, never end-to-end encrypted, became blackmail material.*

BREACHED

THE LOCKS THAT EXIST

- Same Meta 2FA, login alerts and login activity
- "Emails from Instagram" exposes fake security emails
- Restrict and Hidden Words against harassment
- Teen Accounts private by default



HOW TO FIX THIS DOOR

Restore end-to-end encrypted DMs as the default, and blur unsolicited explicit images for all users.

Source: Euronews (2026).

Telegram: the door that only looks locked

MEANWHILE *Nusrat thought her class group on Telegram was private. It isn't: groups are not end-to-end encrypted, and since 2024 IPs and phone numbers can be shared on legal request.*

AT RISK

THE LOCKS THAT EXIST

- Two-step verification cloud password
- Passcode lock and active-session control
- Hide phone number; connect by username
- Secret Chats: end-to-end encrypted, self-destructing

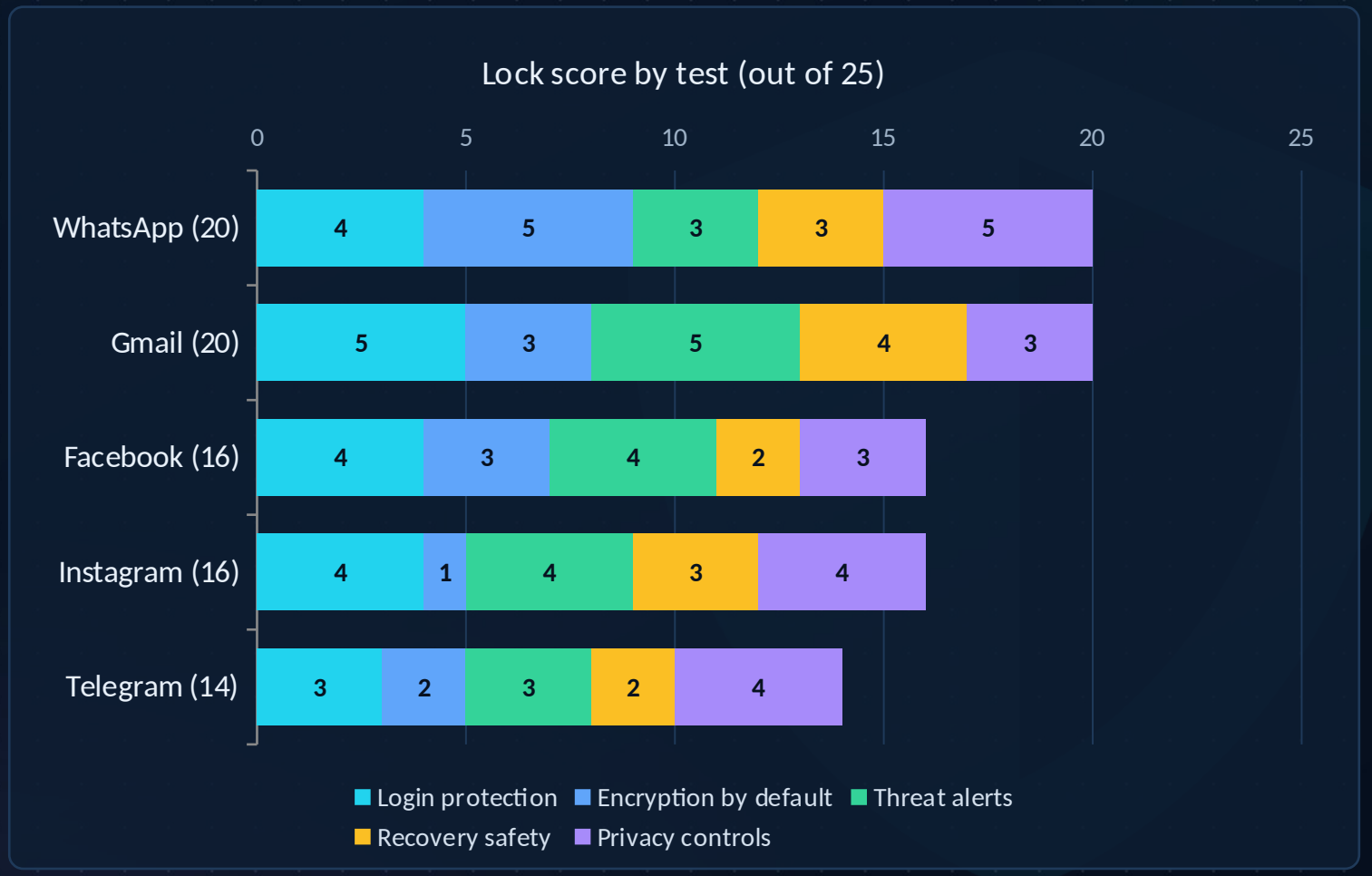


HOW TO FIX THIS DOOR

End-to-end encryption by default for one-to-one chats, and a two-step verification prompt at sign-up.

Source: Engadget (2024).

Which locks held? WhatsApp is the most secure by default



Most secure for individuals: WhatsApp

Strongest lock on by default. It fell only because the owner opened it.

Guard it hardest: Gmail

Once the master key fell, every other door followed.

Least secure: Telegram

Its privacy reputation is stronger than its defaults.

Scores reflect the group's assessment of default and available settings, October 2026.

// 09 REPLAY THE NIGHT

Same night. Different ending.

11:49 PM

Set a WhatsApp two-step PIN

> HIJACK BLOCKED

11:49 PM

Never share a code, even with friends

> DOOR STAYS SHUT

12:20 AM

Unique password and a passkey on Gmail

> MASTER KEY SAFE

12:35 AM

Check recovery email and phone; keep alerts on

> TAKEOVER SPOTTED

01:10 AM

Log out devices you do not recognise

> ATTACKER REMOVED

11:47 PM. The message arrives. ***This time, Nusrat says no.***

Thank you | Questions and discussion